

Hardware data sheet section III addendum

ET1825 / ET1826 / ET1827

Ether**CAT** slave controller
IP core for Lattice® FPGAs

Section I – Technology

(Online at <http://www.beckhoff.com>)

Section II – Register description

(Online at <http://www.beckhoff.com>)

Section III – Hardware description

(Online at <http://www.beckhoff.com>)

Addendum

Design flow compatibility, FPGA device support, known issues

Version 1.1

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BECKHOFF

DOCUMENT ORGANIZATION

The Beckhoff EtherCAT slave controller (ESC) documentation covers the following Beckhoff ESCs:

- ET1200
- ET1100
- EtherCAT IP core for FPGAs

The documentation is organized in three sections. Section I and section II are common for all Beckhoff ESCs, Section III is specific for each ESC variant.

The latest documentation is available at the Beckhoff homepage (<http://www.beckhoff.com>).

Section I – Technology (all ESCs)

Section I deals with the basic EtherCAT technology. Starting with the EtherCAT protocol itself, the frame processing inside EtherCAT slaves is described. The features and interfaces of the physical layer with its two alternatives Ethernet and EBUS are explained afterwards. Finally, the details of the functional units of an ESC like FMMU, SyncManager, distributed clocks, slave information interface, interrupts, watchdogs, and so on, are described.

Since section I is common for all Beckhoff ESCs, it might describe features which are not available in a specific ESC. Refer to the feature details overview in section III of a specific ESC to find out which features are available.

Section II – Register description (all ESCs)

Section II contains detailed information about all ESC registers. This section is also common for all Beckhoff ESCs, thus registers, register bits, or features are described which might not be available in a specific ESC. Refer to the register overview and to the feature details overview in section III of a specific ESC to find out which registers and features are available.

Section III – Hardware description (specific ESC)

Section III is ESC specific and contains detailed information about the ESC features, implemented registers, configuration, interfaces, pinout, usage, electrical and mechanical specification, and so on. Especially the process data interfaces (PDI) supported by the ESC are part of this section.

Additional documentation

Application notes and utilities can also be found at the Beckhoff homepage. Pinout configuration tools for EtherCAT ASICs are available. Additional information on EtherCAT IP cores with latest updates regarding design flow compatibility, FPGA device support and known issues are also available.

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DOCUMENT HISTORY

Version	Comment
1.0	Initial release for EtherCAT IP core 4.0.0
1.1	• Update to EtherCAT IP core for Lattice FPGAs 4.0.0 patch 1 • Update to Lattice Radiant/Propel 2025.2 • Add Certus-N2 support

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1 Overview

This document provides latest release notes, documentation addendum, supported IP core design flows, and supported IP core FPGA types for the following Beckhoff EtherCAT slave controllers:

- EtherCAT IP core for Lattice® FPGAs (up to V4.0.0 patch 1)

Refer to the ESC data sheets for further information. The ESC data sheets are available from the Beckhoff homepage (<http://www.beckhoff.com>).

2 FPGA design tool compatibility

Table 1: EtherCAT IP core for Lattice FPGAs compatibility with Lattice tools

IP core version	Release date	Radiant/Propel version compatibility		
		2024.2	2025.1	2025.2
4.0.0	9/2025	•	•	•
4.0.0 patch 1	1/2026	•	•	•

Table 2: Tool compatibility legend

Symbol	Description
•	Compatible
-	Incompatible
	Not tested

3 FPGA device compatibility

Table 3: EtherCAT IP core for Lattice FPGAs compatibility with Latice FPGAs

IP core version	Avant	Certus-NX	CertusPro-NX	Crosslink-NX	Certus-N2	MachXO5-NX
4.0.0	•	•	•	•	•	•
4.0.0 patch 1	•	•	•	•	•	•

Table 4: FPGA compatibility legend

Symbol	Description
•	Compatible
-	Incompatible
	Not tested

4 Example designs

Table 5: EtherCAT IP core example designs

FPGA board	FPGA	PHY interface	PDI	Processor	Designflow	4.0.0
Lattice Secure Connected Motion Control Platform+I²C EEPROM	CertusPro-NX 100	2x MII	Digital IO	-	Radiant	●
Lattice Secure Connected Motion Control Platform+I²C EEPROM	CertusPro-NX 100	2x MII	AXI	RISC-V	Propel	●

Table 6: Example design availability legend

Symbol	Description
●	Available
	Not available

4.1 Example design compatibility with FPGA design tools

Table 7: Example design compatibility with FPGA tool flow tools

FPGA board	FPGA	PDI	Processor	Example design version	Initial release for IP core	2024.2	2025.1	2025.2
Lattice Secure Connected Motion Control Platform+I²C EEPROM	CertusPro-NX 100	Digital IO	-	1.0	4.0.0	●	●	●
Lattice Secure Connected Motion Control Platform+I²C EEPROM	CertusPro-NX 100	AXI	RISC-V	1.0	4.0.0	●	-	-
Lattice Secure Connected Motion Control Platform+I²C EEPROM	CertusPro-NX 100	AXI	RISC-V	1.1	4.0.0	-	●	●

Table 8: Tool compatibility legend

Symbol	Description
●	Compatible (including resolved issues)
○	Unresolved issues
-	Incompatible
	Not tested

5 Known design flow issues

5.1 Active (at least until Radiant 2025.2)

5.1.1 Radiant 2024.2 – today: timing paths are obfuscated

Radiant uses additional obfuscation for encrypted IP (secured_instance_*), which results in timing paths to be obfuscated. This obfuscation cannot be decoded by Beckhoff, identification of registers and hierarchy inside the EtherCAT IP is not possible.

Solution

Please report critical timing paths to Lattice and request identification.

5.1.2 Radiant 2024.2 – today: assertions and report messages are not evaluated

Radiant does not evaluate assertions and report messages, even if they are based on constants or generics. As a result, messages shown by the EtherCAT IP core source code (text configuration box) are not visible. Additionally, assertions resulting in errors or failures are no longer causing the synthesis to fail, thus, illegal configurations might become synthesized to dysfunctional logic. This is unlikely to happen since the IP core configuration tool prevents illegal configurations, but manual changes might cause them.

Solution

None.

5.1.3 Radiant 2024.2 – today, EtherCAT IP core V4.0.0 before patch 1: User RAM initialization value is byte-swapped for PDI bus width > 8 bit

The user RAM (0x0F80:0x0FFF) initialization values are byte-swapped for 16/32 bit PDI bus width, because the tool flow does not correctly implement the initialization values. Regular RAM access (read/write) during operation is not subject to byte swap.

Solution

Choose one option:

- a) Upgrade to EtherCAT IP core V4.0.0 patch 1
- b) Change the DP_RAM_STYLE configuration in the generated wrapper file from “CODED_BE” to “SIMPLE_BE_WRITE_FIRST”. The IP core configuration tool included in patch 1 does this automatically.
- c) Disable RAM initialization in the IPCore_config tool (integration tab)

6 Troubleshooting

6.1 Licensing issues

6.1.1 Check license status

You can check if your license is valid by invoking Lattice Radiant, menu Help – License Debug...

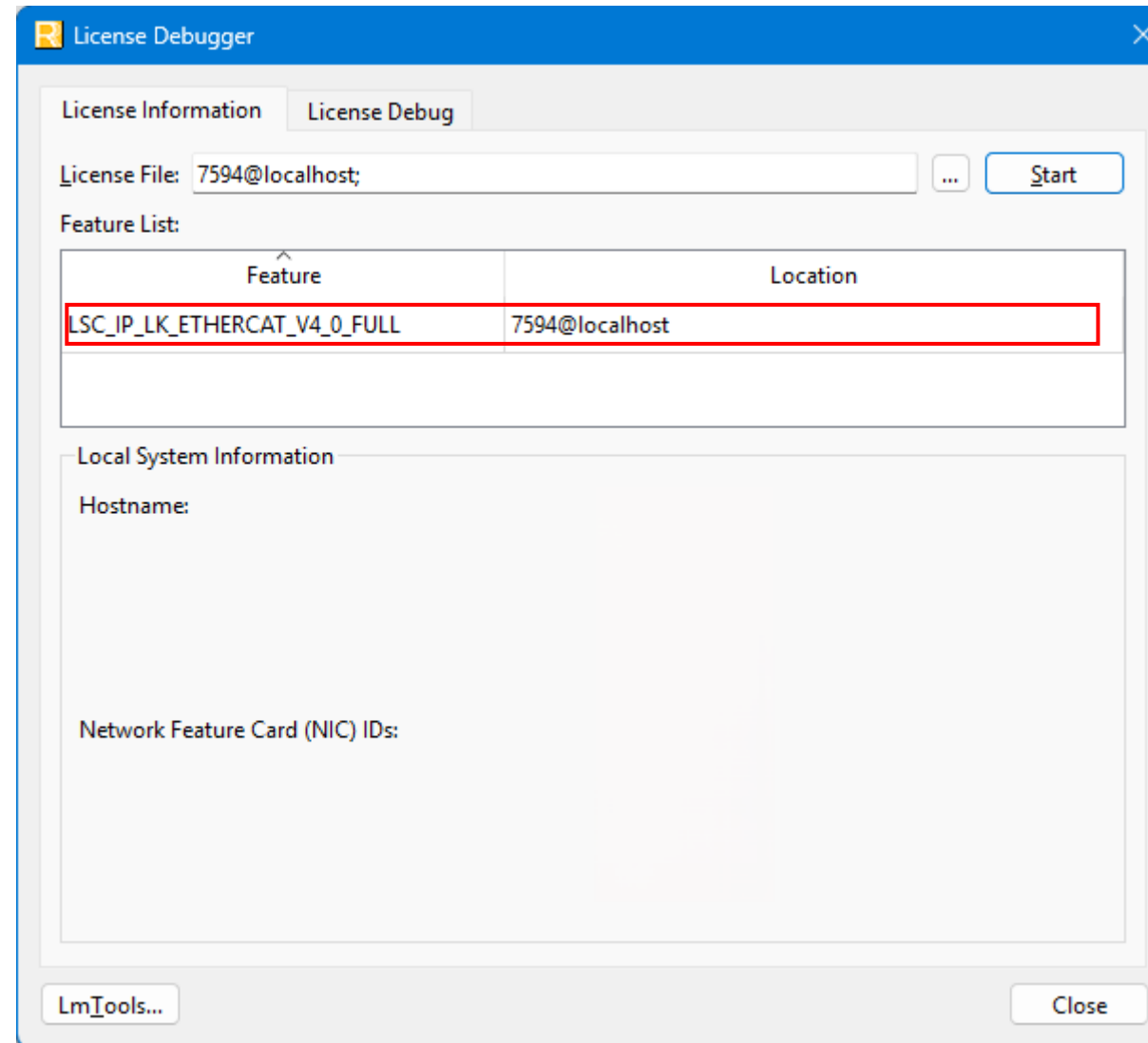


Figure 1: Example license

1. The feature column states the license feature A full license looks like LSC_IP_LK_ETHERCAT_V4_0_FULL, and an evaluation license looks like LSC_IP_LK_ETHERCAT_V4_0_EVAL.
2. The location column shows either the file which contains the node-locked license, or the license server.

NOTE: Additional license information is available using LmTools... in the license debugger.

6.1.2 Check license file entry

If the Lattice Radiant license debugger (menu Help – License Debug) does not report the license as expected, or if you want to find out which license is inside a license file, just open the license file with a text editor. For the EtherCAT IP core, there should be a similar line like this:

Node-locked license

```
FEATURE LSC_IP_LK_ETHERCAT_V4_0_FULL lattice 2026.09 19-sep-2026 uncounted 1234567890AB VENDOR_STRING=LSC_IP_LK_ETHERCAT_V4_0_FULL HOSTID=123456789AB
```

This feature line contains a license with this information:

- It is a license for feature LSC_IP_LK_ETHERCAT_V4_0_FULL, which says it is a feature for the full version of the EtherCAT IP Core. The evaluation version would need LSC_IP_LK_ETHERCAT_V4_0_EVAL.
- The maintenance expiration is 09/2026 (this information is currently not used by the EtherCAT IP).
- The license expires after September 19th, 2026
- The number of seats is uncounted, since it is a node-locked license
- The following hex code and the VENDOR_STRING are interpreted by the license software
- The HOSTID is important, it contains a MAC ID.

Floating license

```
FEATURE LSC_IP_LK_ETHERCAT_V4_0_FULL lattice 2026.03 24-mar-2026 5 1234567890AB VENDOR_STRING=LSC_IP_LK_ETHERCAT_V4_0_EVAL
```

This feature line contains a license with this information:

- It is a license for feature LSC_IP_LK_ETHERCAT_V4_0_FULL, which says it is a feature for the full version of the EtherCAT IP Core. The evaluation version would need LSC_IP_LK_ETHERCAT_V4_0_EVAL.
- The maintenance expiration is 03/2026 (this information is currently not used by the EtherCAT IP).
- The license expires after March 24th, 2026
- The number of seats is 5
- The following hex code and the VENDOR_STRING are interpreted by the license software

With this information, you can compare the license file contents with the license debug information. If the license is not displayed, compare the HOSTID of the license file against the MAC ID shown in the Radiant license debug.

For MAC IDs, make sure the network adapter is enabled. Some adapters enter a power down mode if nothing is connected, try to connect some Ethernet device with the adapter.

6.1.3 No license found

Output log Radiant

```
IP License Checkout failed: LSC IP LK ETHERCAT V4 0 FULL
Failed to checkout 1 LSC IP LK ETHERCAT V4 0 FULL license required for IP
ERROR <2013243> - UI227 |Failed to initialize job: fpga_mapper
```

Message Radiant

```
Error 2013243 Synthesis ERROR <2013243> - UI227 |Failed to initialize job: fpga_mapper
```

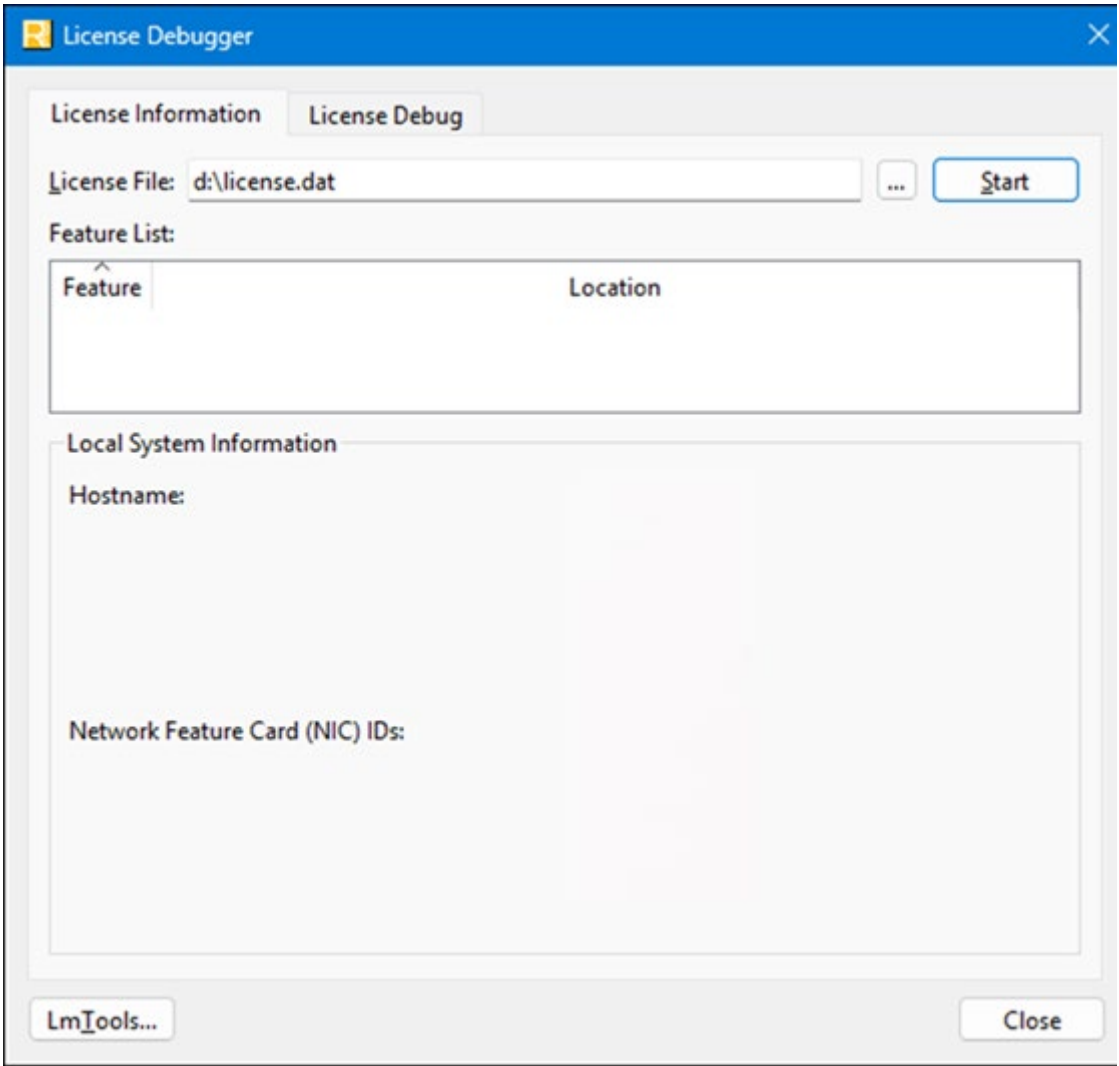


Figure 2: License setup: no license available, expecting “Beckhoff (745C) / 1810”

Solution

Install license according to EtherCAT IP Core datasheet.

6.1.4 No license found, license debug does not show valid license when using remote desktop connection

License debug does not show a valid license using remote desktop, but the license is valid (and shown as valid if user is directly logged into computer).

Output log Radiant

```
IP License Checkout failed: LSC_IP_LK_ETHERCAT_V4_0_FULL
Failed to checkout 1 LSC_IP_LK_ETHERCAT_V4_0_FULL license required for IP
ERROR <2013243> - UI227 |Failed to initialize job: fpga_mapper
```

Solution

Do not use a remote desktop connection, this is not supported.

6.1.5 Evaluation license and IP core identification

Installed license file

If you want to find out if you have the full license or the evaluation license installed, run Lattice Radiant, menu Help – Debug License...

Check the feature line for the EtherCAT IP core:

- LSC_IP_LK_ETHERCAT_V4_0_FULL for the full version (IP core version 4.0.x)
- LSC_IP_LK_ETHERCAT_V4_0_EVAL for the evaluation version (IP core version 4.0.x)

Installed EtherCAT IP core

There are different ways to find out which type of EtherCAT IP core is installed (full/evaluation).

- Remove all EtherCAT IP core licenses and synthesize an example design. The error message gives the type of the license:
“IP License Checkout failed: LSC_IP_LK_ETHERCAT_V4_0_FULL” for the full license, and
“IP License Checkout failed: LSC_IP_LK_ETHERCAT_V4_0_EVAL” for the evaluation license
- Open *EtherCAT_IPCore.vhd* in a text editor and check the comments before the encrypted parts
- Compare the MD5 checksum of *EtherCAT_IPCore.vhd* with the values in the appendix

6.2 Design creation issues

6.2.1 Regenerating EtherCAT IP component fails

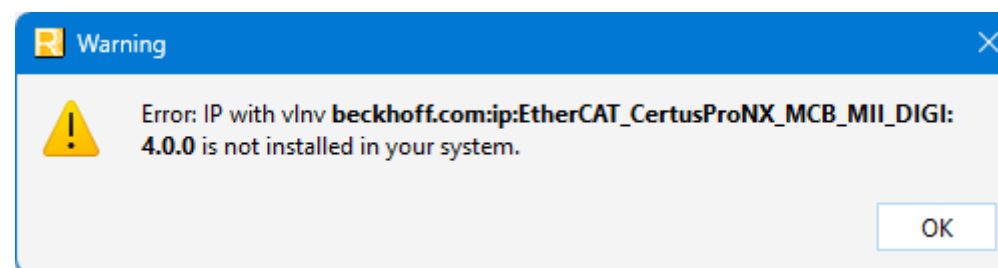


Figure 3: Error message: IP with vlnv beckhoff.com:ip:EtherCAT_CertusProNX_MCB_MII_DIGI:4.0.0 is not installed in your system

Solution

IP generation fails because the configured EtherCAT IP core is not added to the IP catalog. Start EtherCAT IPCore_Config, open the IP configuration file, then press "Generate". Add the updated *.ipk to the IP catalog then regenerate the EtherCAT IP inside Radiant/Propel.

6.3 Synthesis issues

6.3.1 IP component files are missing (example designs)

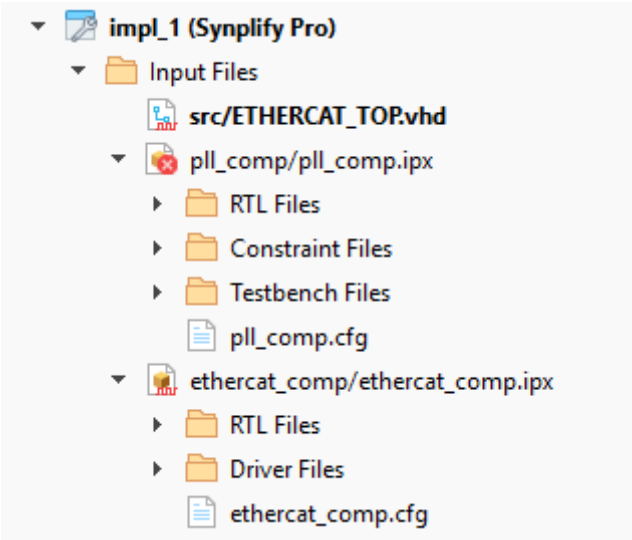


Figure 4: Missing pll_comp files indicated by

Messages

```
Error      35901517 Synthesis ERROR <35901517> - cannot open verilog file pll_comp/rtl/pll_comp.v. VERI-1517
Error      1009990  Synthesis ERROR <1009990> - Stopping Synthesis Tool flow due to error.
Error      35001209 Synthesis ERROR <35001209> - Failed to analyze the file pll_comp/rtl/pll_comp.v.
Error      35001210 Synthesis ERROR <35001210> - Top module pll_comp does not exist in the design.
Error      35001214 Synthesis ERROR <35001214> - Failed to elaborate the design file pll_comp.
Critical   35831027 Synthesis CRITICAL <35831027> - Source compilation failed.
Critical   35831014 Synthesis CRITICAL <35831014> - Error processing input files.
Error      35831010 Synthesis ERROR <35831010> - Processing input files had at least one error. Exiting.
```

Solution

These messages typically appear if you compile an example design for the first time without regenerating the IP components. The errors occur because important component files need to be created. Regenerate all IP components in the design. Compile the project again, the error messages should disappear.

6.3.2 Vendor ID package is missing

Message

```
Warning    35921240 Synthesis WARNING <35921240> - ethercat_ipcore.vhd(0): 'ecat vendor' is not compiled in library work. VHDL-1240 [ethercat_ipcore.vhd:0]
Warning    35921241 Synthesis WARNING <35921241> - ethercat_ipcore.vhd(0): ecat_vendor_id is not declared. VHDL-1241 [ethercat_ipcore.vhd:0]
Warning    2012405  Synthesis WARNING <2012405> - CD643 : "EtherCAT_IPCore.vhd":62139:24:62139:24|Ignoring use clause - ecat_vendor not found ... [EtherCAT_IPCore.vhd:62139]

Error      2012273 Synthesis ERROR <2012273> - CD415 : "EtherCAT_IPCore.vhd":63260:19:63260:19|Expecting keyword 'is' [EtherCAT_IPCore.vhd:63260]
Error      2019992 Synthesis ERROR <2019992> - Parse errors encountered - exiting
```

Solution

Regenerate EtherCAT IP Core, if the vendor ID package is correctly installed, it should be copied to the project folder again.

6.3.3 Important logic parts or I/O signals are optimized away, hardware does not work

If important logic blocks or I/O signals are optimized away, please check if every signal of the EtherCAT IP core is connected, especially the input signals.

6.4 Timing closure issues

6.4.1 General timing closure issues

Regarding constraints, we can only give general advice on increasing the synthesis effort. The default synthesis and implementation features are a good starting point, the following additional options might increase timing closure:

Synthesis Settings

Area = On

General advice

Find out if the violating path consists basically of logic or routing delays. If the logic delays are too high, reduce the number of features, increase optimization effort, or use a faster speed grade.

Generally, the more features are used, the tighter becomes the timing. Especially 64 bit distributed clocks are very demanding, due to the high register count. The low-cost FPGAs are often not able to use 64 bit DC at all. In many cases using 32 bit DC solves the problem.

If many FPGA resources are available, the design is sometimes spread widely across the FPGA, resulting in long connections with high routing delay. Reduce the available FPGA resources/area for the EtherCAT IP core to improve timing.

Generally, it is impossible to predict timing closure for custom user logic in combination with all IP core configurations and FPGA devices. Final certainty can only be reached by implementing the design – that is what the evaluation versions are for. There will be combinations which are not able to reach timing closure at all.

6.5 Hardware issues

If the hardware is not working, the following functions should be tested, e.g., by routing signals to LEDs or using Reveal or external measurement equipment.

- Reset: polarity issues are common, probe exactly the signal which enters the IP core.
- Clock: Are all clocks connected, valid and synchronous (25 MHz and 100 MHz)? Is the PLL locked?
- Turn off MI link detection and configuration for testing if the PHY does not need to be programmed.
- Check in the reports if all FPGA pins are located correctly.
- Is the timing report error-free?

For further information please refer to the data sheet, section I, chapter “troubleshooting and FAQ”.

6.6 Lattice support needs to implement the EtherCAT IP Core for solving an issue

We fully support Lattice in solving customer issues. We provide EtherCAT IP cores, licenses, and vendor ID packages containing the Lattice vendor ID directly to Lattice.

7 Appendix

7.1 MD5 checksums

Table 9: MD5 checksums

IP core version	ZIP file (eval)		EtherCAT_IPCore.vhd (eval)	EtherCAT_IPCore.vhd (full)
4.0.0	ZIP eval:	E5FCDF6184B1E8C333FBCA2FEF6D3E4D	205ACF26498FFE9719BBA1C729CCA248	61125765E7BE7A62526CAC3E50E20D08
	ZIP full:	30B1F5EF01FF6BFFD850B9B1EF25F7AA		
4.0.0 patch 1	ZIP eval:	A13DD6B85A0E99632788D536E6AEBB0D	205ACF26498FFE9719BBA1C729CCA248	61125765E7BE7A62526CAC3E50E20D08
	ZIP full:	5A84D6551BDA50CE9950DD6E76D498AE		

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The addresses of Beckhoff's branch offices and representatives round the world can be found on her internet pages: <http://www.beckhoff.com>

You will also find further documentation for Beckhoff components there.

7.4 Beckhoff headquarters

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